

SEATTLE HARBOR, WA, USA

WMO: 994014

Lat: 47.6026N

Lon: 122.3392W

Elev: 7

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 05-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 29.0	(c) 32.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 20.3	(k) 49.9	(l) 16.8	(m) 46.7	(n) 4.4	(o) 130	(p) 0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.8	(c) 79.3	(d) N/A	(e) 76.0	(f) N/A	(g) 73.0	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 6.5	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.3	(b) 13.5	(c) 11.6	(e) 24.1	(f) 87.1	(g) 3.5	(h) 4.0	(i) 21.6	(j) 90.0	(k) 19.5	(l) 92.4	(m) 17.5	(n) 94.6	(o) 15.0	(p) 97.6		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.5	44.0	44.7	47.6	50.9	56.5	60.5	64.9	65.0	61.5	54.7	47.4	43.1
	DBStd	9.04	5.06	4.88	4.36	4.14	4.61	4.18	3.85	3.17	3.76	3.91	5.35	5.66
	HDD50	823	194	152	101	35	2	0	0	0	0	8	109	222
	HDD65	4325	652	567	540	423	267	147	49	40	116	318	528	678
	CDD50	2084	7	5	25	62	204	314	462	465	346	155	31	8
	CDD65	114	0	0	0	0	4	12	46	40	12	0	0	0
	CDH74	504	0	0	0	2	34	74	217	145	32	0	0	0
	CDH80	81	0	0	0	0	7	10	43	18	3	0	0	0
Wind	WSAvg	5.0	5.2	5.2	6.0	5.8	5.1	4.9	4.4	4.1	4.3	4.5	5.3	5.0
Precipitation	PrecAvg	38.8	5.2	4.0	3.9	2.9	1.9	1.6	1.0	1.3	1.9	3.3	5.6	6.2
	PrecMax	54.5	10.4	8.6	8.7	5.8	4.3	4.1	4.3	5.5	5.6	8.0	11.2	12.0
	PrecMin	27.8	0.6	0.6	0.4	0.9	0.4	0.1	0.0	0.0	0.0	0.3	0.5	1.2
	PrecStd	6.0	2.3	1.8	1.7	1.2	0.9	1.0	0.8	1.2	1.4	1.9	2.2	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.9	58.1	63.9	69.5	78.2	81.2	85.5	82.0	77.9	68.5	61.7	59.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	54.4	55.0	58.7	63.8	71.4	75.7	79.8	77.8	73.5	64.3	58.3	54.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	52.4	52.8	55.9	59.7	66.8	70.9	75.9	74.6	70.2	62.2	56.6	52.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	50.4	51.0	53.8	57.0	63.6	67.6	72.4	71.6	67.6	60.2	54.7	50.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	7.2	8.2	10.0	11.5	12.4	12.5	14.8	14.4	11.6	8.6	8.0	7.0
		MCDBR	8.4	10.5	13.0	16.0	18.5	18.5	21.1	19.6	16.4	11.3	8.3	8.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.328	0.318	0.327	0.358	0.377	0.366	0.353	0.359	0.347	0.332	0.323	0.332
	taud		2.488	2.548	2.541	2.426	2.392	2.447	2.503	2.495	2.519	2.564	2.539	2.410
	Ebn at Noon		239	272	285	284	281	285	287	281	274	262	240	219
	Edh at Noon		21	23	28	34	36	35	32	31	28	23	19	20
All-Sky Solar Radiation	RadAvg		319	597	900	1325	1671	1835	2048	1745	1267	715	374	259
	RadStd		45	79	103	123	135	181	152	116	113	78	50	37

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page